
Curation Metrics Are a Post-Training Decision: Auditing Trajectory Smoothness for Adapting Robot Foundation Models

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Abstract

1 Trajectory smoothness has been proposed as a cheap, training-free signal for curat-
2 ing imitation-learning demonstrations, most recently by RINSE, which introduces
3 Spectral Arc Length (SAL) and a contact-aware Trajectory-Envelope Distance
4 (TED) and reports downstream gains on robomimic. Recent audit work showed
5 that generic curation metrics on a synthetic-defect LIBERO testbed largely ex-
6 ploit episode length rather than demonstration content. We test whether the
7 *specific published metrics* survive the corresponding controls on *real* data. On
8 robomimic can/mh, SAL separates better from worse operators with AUROC
9 0.952, but episode length alone achieves 0.957, and inside the tightest duration-
10 matched band SAL falls to 0.628 while raw mean jerk, the baseline RINSE argues
11 against, rises to 0.896. On can/mg, where every episode is exactly 150 steps and
12 length cannot carry information, SAL is at chance (0.454) and TED is inverted
13 (0.376) against the task-success label, while jerk reaches 0.706. Training GMM
14 behavior-cloning policies on each metric’s curated subset (10 seeds per condi-
15 tion), curating by TED costs roughly a third of downstream success relative to
16 random selection of the same size (0.240 vs. 0.360, $p=0.0001$, surviving Bonfer-
17 roni correction); path length is directionally harmful with weaker support, while
18 SAL and jerk are not statistically distinguishable from not curating at all. In-
19 specting individual trajectories shows why: on this task successful manipulation
20 requires sustained high-deviation motion through the contact phase, while failure
21 manifests as smooth low-amplitude hovering, so a smoothness metric measures
22 commitment with the sign flipped. The length confound replicates on a second
23 task: on square/mh, SAL correlates +0.96 with episode duration and its skill AU-
24 ROC falls from 0.898 to 0.480 inside a duration-matched band, while jerk rises
25 from 0.431 to 0.896. For post-training pipelines that filter, reweight, or select
26 demonstrations before fine-tuning a pretrained policy, the practical consequence
27 is direct: the curation step can silently discard the corrective, high-effort demon-
28 strations that fine-tuning most needs, while keeping the smooth, low-information
29 ones. Our results extend the detection-vs-curation decoupling to named, published
30 metrics, on operator labels rather than injected defects, and to a benchmark where
31 the length confound is excluded by construction rather than by truncation.

1 Introduction

33 A large share of post-training work for robot foundation models runs on curated demonstration sub-
34 sets: filtering, reweighting, or ranking trajectories before fine-tuning, whether the source is human
35 teleoperation, autonomous rollouts, or a mix of both. The curation step is rarely evaluated as care-

fully as the fine-tuning step that follows it, and the metrics used to perform it are typically validated once, on clean data, and then reused across deployments without re-checking whether they still measure what they claim to measure. This paper is a case study in what goes wrong when that assumption fails: a published, named curation metric that looks reliable on successful demonstrations, and is actively harmful once the pool it filters contains the failures and near-misses that post-training adaptation pipelines actually have to work with.

Behavior cloning inherits the quality of its demonstrations, and a growing literature proposes to score and filter demonstrations automatically. Training-free trajectory statistics are the cheapest family of such scores, and the most developed recent proposal is RINSE [Kulkarni et al., 2026], which ranks demonstrations by two smoothness metrics: Spectral Arc Length (SAL), a frequency-domain measure adapted from rehabilitation science, and Trajectory-Envelope Distance (TED), a contact-aware spatial deviation from a Bézier-smoothed reference. RINSE reports that filtering by these metrics matches or beats full-data training on robomimic tasks.

A recent line of audit work complicates this picture. Bedi [Bedi, 2026a] showed on a controlled testbed that action-only curation scorers are blind or inverted on structural defects, and the follow-up [Bedi, 2026b] showed that five of seven curation metrics on a LIBERO testbed achieved near-perfect detection AUROC by measuring episode length rather than demonstration content, and that detection accuracy and curation value are sharply decoupled. Those results, however, were obtained with *generic* metric implementations (a SPARC-style smoothness score among them), with *synthetically injected* defects, and with episode length controlled by *truncation*.

This paper asks the natural next question: do the specific, published metrics of RINSE survive these controls on real data? Our audit differs from the Bedi testbeds on three axes, each of which closes a gap that line of work explicitly left open:

1. **Named metrics, not proxies.** We implement SAL from its published formula and TED from the RINSE appendix, including the contact-boundary partition that is TED’s distinguishing design feature. TED has not previously been evaluated outside its authors’ own success-filtered setting.
2. **Real labels, not injected defects.** robomimic can/mh [Mandlekar et al., 2021] carries *human operator skill tiers* (better/okay/worse) assigned to real teleoperators, and can/mg carries real task-success outcomes of a scripted-policy data-generation process. No defect is synthetic.
3. **A length-free benchmark by construction.** In can/mg every episode is exactly 150 steps. The episode-length confound documented by [Bedi, 2026b] cannot exist in this data, with no truncation required. Truncation removes the confound by discarding signal; a fixed-horizon dataset removes it without touching the trajectories.

Our findings are easy to state. Smoothness metrics rank *operator skill* largely to the extent that skill is confounded with how long the episode took. Once duration is controlled, by normalization or by duration-matched subsets, SAL’s advantage collapses and raw mean jerk, the crude baseline the smoothness literature positions itself against, becomes the strongest detector. Against *task success*, on fixed-length data, both of RINSE’s metrics are at or below chance, with TED inverted, while jerk retains signal. Curating by TED produces a policy substantially worse than random subsampling of the same size, a result that survives correction for multiple comparisons; the weaker metrics are not distinguishable from doing nothing. Skill and success are different targets, and the curation literature validates on one while deploying against the other.

2 Related Work

Trajectory-statistic curation. RINSE [Kulkarni et al., 2026] is the primary subject of this audit; it assumes demonstration pools pre-filtered for task success and validates on successful trajectories only, a scoping decision its limitations section states explicitly. PSD [Sojib and Begum, 2026] scores demonstrations by total spectral power of the end-effector trajectory and reports gains over learning-based curation on robomimic-MH and a non-expert dataset. Neither work evaluates on pools containing failures.

Audits of curation metrics. Bedi’s testbed papers [Bedi, 2026a,b,c] established the detection-vs-curation decoupling, documented the episode-length confound, and reported a negative result on per-phase metric selection. We adopt their framing and extend it to published metrics, real labels, and a natively fixed-length benchmark. Where they control length by truncating to a fixed prefix, our can/mg experiments require no such intervention.

Evaluation practice. Kress-Gazit et al. [Kress-Gazit et al., 2024] and RoboEval [Wang et al., 2026] argue that binary success rate under-describes policy quality, and [Jiang et al., 2026] shows that widely used benchmarks admit shortcut solutions. Our results are a data-side complement: the scores used to *select training data* can also be shortcut-driven, with episode duration as the shortcut.

3 Metrics and Datasets

Metrics. All metrics are computed from proprioceptive signals at the robomimic control rate of 20 Hz ($\Delta t = 0.05$ s) and return a badness score (higher is worse). *SAL*: spectral arc length of the end-effector speed profile, implemented from the published formula; we negate so that higher means less smooth. *TED*: contact-aware trajectory-envelope distance, reimplemented from the RINSE appendix: the 6-D pose trajectory (position plus rotation vector, orientation weight $w_{\text{ori}} = 0.25$) is partitioned at gripper open/close transitions, each segment is fit with a greedy Bézier envelope, and the score is the length-normalized DTW distance between trajectory and envelope. *Jerk*: mean third-difference magnitude of end-effector position. *Path length*: total end-effector arc length. Where indicated we also report per-timestep normalizations (score divided by $T-1$). Our TED is a reimplementaion from the paper, not the authors’ code; Section 7 discusses the resulting caveat and the sanity check we ran.

Datasets. All data is public robomimic [Mandlekar et al., 2021] low-dim. *can/mh*: 300 successful demonstrations from six human operators with released skill tiers (better/okay/worse, 100 each); episode lengths range 98–1050 steps. *can/mg*: 3900 machine-generated demonstrations, 18.4% successful, every episode exactly 150 steps. *can/paired*: 200 demonstrations in matched success/failure pairs (50% successful), lengths 55–147; we report it as a secondary setting because its matched-pair construction makes it adversarially hard for trajectory-level scores. *square/mh*: 300 successful demonstrations with the same skill-tier structure (100 each), episode lengths 123–1051, used in Section 4.4 to test whether the length confound is task-specific. We note that Can is the only robomimic task with failure-containing splits (mg and paired); Square, Transport, and Tool Hang ship only proficient-human and multi-human data, all of it successful. Our success-label experiments are therefore confined to Can by benchmark availability, not by choice.

Statistics. All AUROCs are reported with 95% bootstrap confidence intervals (2000 resamples). Downstream comparisons use difference-of-means bootstrap CIs (20,000 resamples) and Welch two-sample tests, with Bonferroni correction applied across the full set of reported comparisons; we mark which results survive correction and which do not. For skill we report better-vs-worse AUROC; for success we report AUROC against the binary success label with scores negated so that $\text{AUROC} > 0.5$ means the metric ranks clean/successful demonstrations higher, and $\text{AUROC} < 0.5$ means the metric is *inverted*.

4 Detection Results

4.1 Operator skill on can/mh: the length confound, on real labels

On the full 200 better/worse demonstrations, SAL appears excellent: AUROC 0.952 [0.922, 0.975]. But episode length alone achieves 0.957 [0.929, 0.980], and SAL’s correlation with T is +0.97. The tier means make the mechanism plain: worse operators average 304 steps, better operators 143, and SAL’s tier means (100.4 / 57.5 / 44.9) track duration almost linearly. On this dataset SAL is, to first order, a stopwatch.

Two independent controls confirm this. First, *normalization*: SAL’s arc length sums over $T/2$ spectral bins, so dividing by $T-1$ is the principled per-timestep correction; normalized SAL drops from 0.952 to 0.582 [0.497, 0.662] and its length correlation drops from +0.97 to +0.07. Second, *duration matching*: restricting to bands of increasingly similar episode length (Figure 1a, Table 1). In the tightest band ($T \in [140, 175]$, $n=40$), where length itself is at chance (0.485 [0.227, 0.740]), SAL

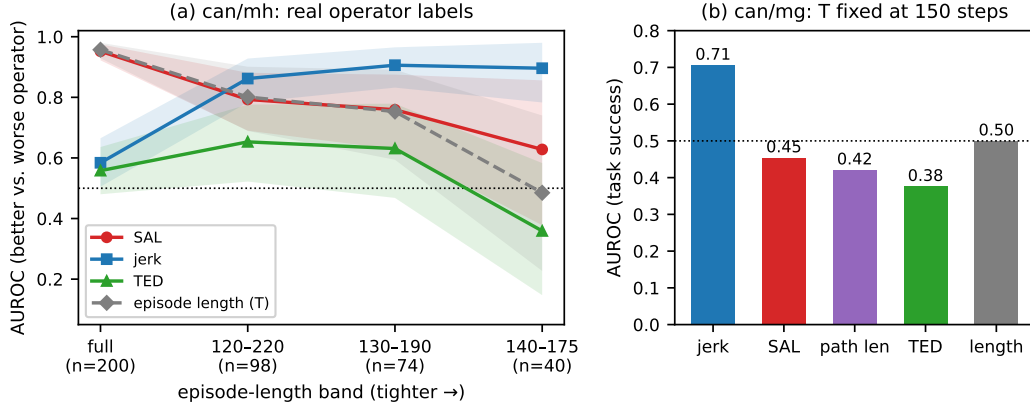


Figure 1: **Smoothness metrics rank duration, not quality.** (a) Better-vs-worse operator AUROC on can/mh across duration bands of increasing tightness (shaded: 95% bootstrap CI). SAL (red) tracks the episode-length baseline (gray, dashed) almost exactly and falls from 0.952 to 0.628 as the band tightens; raw jerk (blue) moves the opposite way, 0.584 to 0.896; TED (green) never clears chance and inverts in the tightest band. In that band episode length itself is at chance (0.485), confirming the control. (b) On can/mg every episode is exactly 150 steps, so length carries no information by construction; against the task-success label, jerk retains signal (0.706), SAL is at chance (0.454), and TED is inverted (0.376).

	full (n=200)	$T \in [120, 220]$ (n=98)	$T \in [130, 190]$ (n=74)	$T \in [140, 175]$ (n=40)
SAL	0.952 [.922,.975]	0.793 [.690,.881]	0.759 [.628,.875]	0.628 [.378,.856]
SAL/ T	0.582 [.497,.662]	0.600 [.479,.715]	0.603 [.446,.755]	0.658 [.342,.923]
TED	0.558 [.480,.636]	0.653 [.522,.776]	0.631 [.468,.779]	0.359 [.147,.583]
jerk	0.584 [.505,.665]	0.862 [.783,.928]	0.906 [.832,.965]	0.896 [.783,.980]
path len.	0.783 [.718,.842]	0.590 [.449,.730]	0.624 [.461,.781]	0.429 [.215,.645]
length T	0.957 [.929,.980]	0.801 [.689,.901]	0.753 [.594,.891]	0.485 [.227,.740]

Table 1: Better-vs-worse operator AUROC on can/mh across duration bands, with 95% bootstrap CIs. The bottom row is the control: a band is duration-matched only where T itself is near chance. Per-timestep normalizations of TED and path length are omitted: unlike SAL, whose raw value grows with the number of spectral bins, these quantities do not scale linearly with T , and dividing by T turns them into inverse-speed proxies rather than duration-free measures of smoothness.

is 0.628 [0.378, 0.856], a CI that includes chance, while raw jerk is 0.896 [0.783, 0.980]. Across the four bands the two metrics *cross*: as the confound is removed, the crude metric strengthens monotonically (0.584, 0.862, 0.906, 0.896) and the sophisticated one weakens monotonically (0.952, 0.793, 0.759, 0.628).

TED never clears chance in any band (0.558 full-sample, upper CI 0.636) and inverts to 0.359 in the tightest band. TED does *not* have the length pathology (correlation +0.14 with T); it simply carries little skill signal on this data.

We caution that the tightest band retains only 7 worse-operator demonstrations and its intervals are wide; the claim rests on the *trend* across all four bands and its agreement with the normalization control, not on any single cell. We also note that naïve per-timestep normalization is only appropriate for metrics whose raw value grows with T ; normalized path length, for example, becomes an inverse-speed proxy that *rises* with skill and is reported only for completeness in Table 1.

4.2 Task success on can/mg: the confound removed by construction

The can/mg split fixes every episode at 150 steps, so no metric can exploit duration. Against the success label (716 of 3900 successful), jerk achieves 0.706, SAL 0.454, path length 0.421, and TED 0.376 (Figure 1b). The length baseline is exactly 0.500, as it must be. Per-timestep normaliza-

	full ($n=200$)	$T \in [150, 300]$ ($n=111$)	$T \in [180, 260]$ ($n=55$)	$T \in [200, 240]$ ($n=30$)
SAL	0.898 [.852,.936]	0.759 [.658,.849]	0.529 [.344,.702]	0.480 [.250,.701]
SAL/ T	0.458 [.376,.536]	0.485 [.372,.602]	0.468 [.300,.653]	0.520 [.280,.738]
TED	0.663 [.584,.736]	0.685 [.577,.788]	0.599 [.427,.758]	0.557 [.335,.785]
jerk	0.431 [.349,.509]	0.570 [.455,.687]	0.782 [.618,.923]	0.896 [.724,1.000]
path len.	0.931 [.893,.963]	0.817 [.726,.895]	0.796 [.654,.913]	0.887 [.742,.991]
length T	0.936 [.902,.964]	0.824 [.738,.898]	0.613 [.456,.766]	0.407 [.197,.629]

Table 2: Better-vs-worse operator AUROC on *square/mh* across duration bands, with 95% bootstrap CIs, in the same format as Table 1. The bottom row is the control. The SAL/jerk crossover observed on Can reproduces here: SAL tracks the length baseline down while jerk rises, and in the band where length is at chance jerk reaches the same 0.896 it reaches in Can’s tightest band.

tion leaves every AUROC unchanged, as expected under a constant divisor, which we report as a consistency check.

Two readings of this result matter. First, both of RINSE’s metrics are at or below chance on the label practitioners actually deploy against. TED’s inversion means that filtering by TED *preferentially retains failed demonstrations*. Second, the ordering among metrics reverses relative to the uncontrolled skill setting: the metric that looked worst there (jerk) is the only one with signal here.

4.3 Secondary setting: can/paired

On the matched-pair dataset every metric is inverted against success (TED 0.331, SAL 0.398, jerk 0.368, path length 0.029), but the setting is confounded in the opposite direction: successful demonstrations are systematically *longer* (length-alone AUROC 0.194), and path length’s +0.86 correlation with T makes it a pure length detector here. We report can/paired for completeness and treat can/mg as the primary success-label evidence.

4.4 Generalization to a second task: square/mh

If the length confound were an artifact of the Can task, the argument would be considerably weaker. We therefore repeat the skill-label analysis on square/mh, a harder insertion task with the same operator skill-tier structure and a comparable duration range (123–1051 steps versus Can’s 98–1050). No retuning was performed: the same metric implementations and the same band procedure were applied to the new data.

The confound replicates almost exactly. SAL’s correlation with episode duration is +0.96 on Square against +0.97 on Can, and its uncontrolled skill AUROC is 0.898 [0.852, 0.936] against a length-alone baseline of 0.936 [0.902, 0.964]. Per-timestep normalization collapses it to 0.458 [0.376, 0.536], slightly below chance and a larger drop than the corresponding Can figure. Path length behaves the same way, reaching 0.931 [0.893, 0.963] uncontrolled with a +0.90 duration correlation.

The crossover also replicates (Table 2). As the duration band tightens and the length baseline falls from 0.936 to 0.407, SAL falls monotonically (0.898, 0.759, 0.529, 0.480) while jerk rises monotonically (0.431, 0.570, 0.782, 0.896). In the band where duration is genuinely controlled ($T \in [200, 240]$, length AUROC 0.407), jerk reaches 0.896 [0.724, 1.000], numerically identical to its value in Can’s tightest band, while SAL sits at 0.480 [0.250, 0.701], a CI centred on chance.

Two differences from Can are worth stating. First, jerk is *inverted* on Square before any control is applied (0.431), where on Can it was weakly positive (0.584); the uncontrolled ranking of metrics is therefore itself task-dependent, even though the controlled ranking is not. Second, path length remains strong in the controlled band on Square (0.887 [0.742, 0.991]) where it collapsed on Can (0.429 [0.215, 0.645]). We do not claim jerk or path length is a good curation metric on the strength of these numbers; the point is narrower, that whatever advantage SAL displays over them disappears once duration is held fixed, on both tasks. The tightest Square band retains 13 worse-operator demonstrations, so as with Can the claim rests on the monotone trend across bands rather than on any single cell.

condition	pool size	best success rate (mean $\pm$ std)	95% CI
oracle (successes only)	716	0.892 ± 0.048	[0.866, 0.922]
no curation	1950	0.380 ± 0.063	[0.344, 0.420]
random 50%	1950	0.360 ± 0.061	[0.326, 0.398]
SAL	1950	0.332 ± 0.061	[0.298, 0.370]
jerk	1950	0.332 ± 0.041	[0.306, 0.354]
path length	1950	0.310 ± 0.030	[0.292, 0.328]
TED	1950	0.240 ± 0.045	[0.212, 0.266]

Table 3: Downstream GMM-BC policy success on can/mg after filtering to a fixed 1950-demonstration pool by each metric, 10 seeds each. The oracle pool is smaller (716, the number of successes that exist) and is a ceiling reference, not a matched condition. Pairwise tests against the baselines are in Table 4.

metric	vs. random 50%		vs. no curation	
	Δ [95% CI]	p	Δ [95% CI]	p
TED	-0.120 [$-0.166, -0.076$]	0.0001	-0.140 [$-0.186, -0.096$]	< 0.0001
path length	-0.050 [$-0.090, -0.012$]	0.037	-0.070 [$-0.112, -0.030$]	0.0071
jerk	-0.028 [$-0.072, +0.014$]	0.25	-0.048 [$-0.092, -0.006$]	0.060
SAL	-0.028 [$-0.080, +0.024$]	0.32	-0.048 [$-0.098, +0.002$]	0.098

Table 4: Difference in mean best success rate between each metric-curated condition and the two baselines, with 95% bootstrap CIs on the difference (20,000 resamples) and Welch two-sample p -values. Bold marks the comparisons surviving Bonferroni correction across all eight tests ($\alpha = 0.05/8 = 0.006$). Only TED clears that bar. For reference, random and no-curation are themselves indistinguishable ($\Delta = -0.020$, CI [$-0.072, +0.032$]).

5 Does Detection Predict Curation? Downstream Training

Detection AUROC is not the quantity practitioners care about [Bedi, 2026b]; the trained policy is. We therefore filter can/mg by each metric at a 50% keep fraction and train a GMM behavior-cloning policy (MLP 1024×1024 , 5 modes, batch 100, 600 epochs of 500 gradient steps, learning rate 10^{-4}) on each curated subset, with 50 evaluation rollouts (horizon 400) every 150 epochs and the best checkpoint reported per seed. Conditions: TED, SAL, jerk, path length, plus three references: an *oracle* keeping only successful demonstrations, a *random 50%* subset, and an uncurated size-matched control. Five seeds per condition. All filtered pools contain 1950 demonstrations except the oracle (716, the number of successes that exist); we therefore treat the oracle as a ceiling reference rather than a matched condition.

Table 3 answers the question Section 4 could only pose, and Table 4 reports the comparisons that matter. Two results are unambiguous. Curating by TED costs roughly a third of downstream success relative to random selection of the same size (-0.120 , $p = 0.0001$) and relative to not curating at all (-0.140 , $p < 0.0001$); both survive Bonferroni correction across the eight comparisons we report. Path length is directionally harmful with weaker support ($p = 0.037$ and $p = 0.0071$), clearing conventional significance but not correction. SAL and jerk sit below both baselines in the mean but are not statistically distinguishable from them ($p \geq 0.06$), and we report them as such rather than as evidence of harm.

We emphasize this because an earlier version of this analysis, run with 5 seeds rather than 10, appeared to show every metric-curated condition underperforming both baselines. That pattern did not survive the seed expansion for SAL and jerk. Since the argument of this paper is that underpowered evaluation produces confident claims in the wrong direction, we apply the same standard to our own results: the defensible claim is about TED, not about smoothness metrics as a class.

What the grid does establish irrespective of any single comparison is that none of these metrics recovers usable signal that is demonstrably present. An oracle keeping only true successes reaches 0.892 [0.866, 0.922], roughly $2.5\times$ the best metric-curated condition, so the pool contains ample information about which demonstrations to keep; the trajectory statistics simply do not find it. Ran-

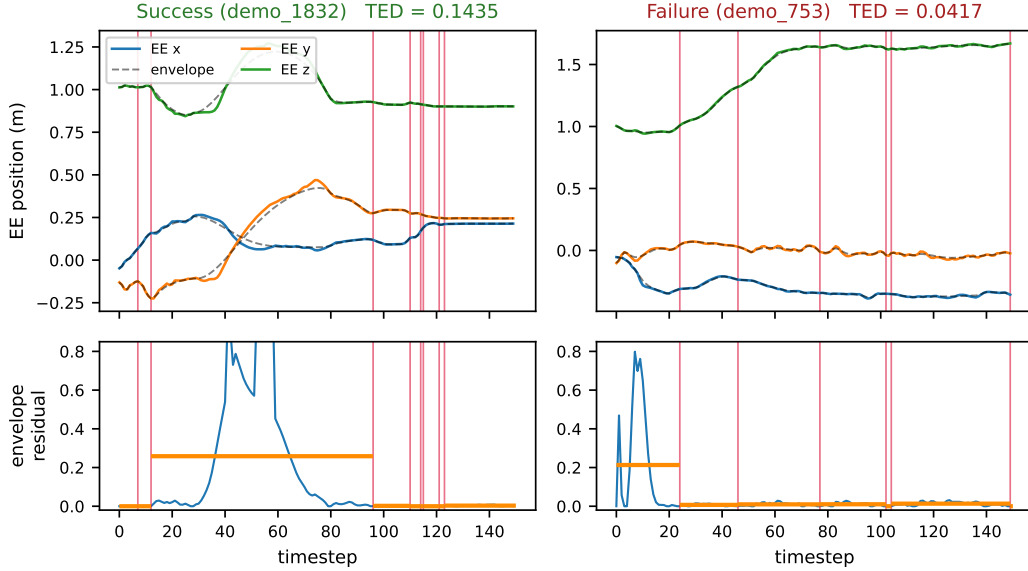


Figure 2: **Failure is smooth; success is not.** A successful (left) and failed (right) can/mg demonstration, with end-effector position, TED’s gripper-contact segment boundaries (red), the per-segment Bézier envelope (dashed), and the per-timestep envelope residual (bottom, orange bars are per-segment means). Both trajectories contain one high-deviation segment, but in the success it spans timesteps 12–96 and coincides with the descend-and-grasp, while in the failure it is confined to the initial free-space approach (0–24) and the remaining 126 steps are a flat tail of repeated gripper cycles that never completes a placement. TED averages over the alignment path and so rates the failure as the cleaner trajectory. Illustrative of the mechanism only; the quantitative claim is Figure 1b.

dom subsampling and the uncurated control are themselves indistinguishable ($\Delta = -0.020$, CI spanning zero), which is the sanity check one wants: at a 50% keep fraction on this data, discarding demonstrations at random costs nothing measurable, so any deficit a metric shows is attributable to *which* demonstrations it discards rather than to the reduction in pool size.

The downstream ordering tracks the detection ordering imperfectly. TED is worst in both, and path length is weak in both, but jerk has the best detection AUROC on can/mg (0.706) while landing indistinguishable from SAL downstream, whose detection AUROC is near chance (0.454). Detection accuracy is at best a coarse predictor of curation value, consistent with the decoupling reported by [Bedi, 2026b] on a different testbed.

6 Discussion

Skill and success are different targets. The same metric family gives opposite verdicts depending on the label. Under an uncontrolled skill label, SAL looks nearly perfect; under a length-controlled skill label it is marginal; under a success label on fixed-length data it is at chance and TED is inverted. Curation papers validate against operator skill or smoothness-adjacent labels on success-only pools, then are deployed, in practice, to remove *failures* from mixed pools. Our results say those are not the same job, and the metrics do not transfer between them.

The crude baseline wins under controls. RINSE motivates SAL and TED as principled improvements over naïve statistics such as jerk. Under every length control we applied, that ordering reverses. We do not claim jerk is a good curation metric in general, and the Bedi testbeds [Bedi, 2026a] show smoothness-family scorers failing badly on structural defects; we claim only that on real robomimic data the published sophistication does not survive the controls that the audit literature has shown to be necessary.

Why TED inverts: failure is smooth. TED’s design premise is that partitioning at gripper-contact boundaries isolates task-relevant abruptness, and a plausible hypothesis going in was that this would

let TED catch failures, since canonical structural failures are contact-boundary events. The data supports a different and simpler mechanism. Figure 2 contrasts a successful and a failed can/mg demonstration. Both contain one high-deviation segment, but they differ in when it occurs and how much of the episode it spans. In the success, the high-residual segment runs from timestep 12 to 96 (84 of 150 steps, mean residual 0.259) and coincides with the descend-and-grasp: the end effector dips, the lateral axes swing, and the envelope cannot track the motion. In the failure, the only comparable deviation is the initial free-space approach (timesteps 0–24, mean residual 0.213); after that the trajectory settles into a flat tail in which the end effector holds height while the gripper opens and closes repeatedly without completing a placement, and every remaining segment has residual below 0.014. Because TED averages deviation over the DTW alignment path, a committed motion spanning half the episode raises the score substantially while a short approach transient is diluted by the smooth remainder. TED therefore rates the failure as the cleaner trajectory (0.042 vs. 0.144).

On this task, in other words, successful manipulation *requires* sustained high-deviation motion through the contact phase, while failure manifests as smooth hovering rather than as erratic thrashing. A metric that rewards smoothness is measuring commitment with the sign flipped. This also explains why raw jerk is the one metric that retains signal in Section 4: jerk is high precisely in the trajectories that commit. We stress that these panels illustrate a mechanism rather than establish it; the quantitative claim is the AUROC of 0.376 in Figure 1b, and the per-demonstration TED scores we inspected (successes 0.144 and 0.079, failures 0.042, 0.042, and 0.017) are consistent with but not evidence for that number.

7 Limitations

Our detection results cover two robomimic tasks (Can and Square) but our success-label and downstream-training results cover only Can, because Can is the only robomimic task shipping failure-containing splits; extending the success-label analysis requires moving to a different benchmark such as LIBERO. We use one policy class (GMM behavior cloning); the detection results are policy-free but the downstream results may differ under sequence models or diffusion policies. We piloted a diffusion-policy replication of the downstream grid on a subset of conditions (TED, jerk, random, no-curation, oracle; 3 seeds) under a reduced-compute configuration (300 vs. 600 training epochs, 30 vs. 50 evaluation rollouts, 10-step DDIM inference) adopted to keep wall-clock time tractable. Absolute success rates were substantially lower than the BC grid across every condition, including the oracle (0.489 vs. 0.892), indicating the reduced configuration had not reached a comparable operating point rather than reflecting a property of the architecture. We do not consider this pilot adequately powered or matched to report as a finding, and do not include its numbers above; a diffusion-policy replication under compute matched to the BC grid is the clearest piece of follow-up work. Our TED is a reimplement from the RINSE appendix, not the authors’ released code; as a sanity check it broadly agrees with SAL’s ranking on the clean proficient-human data, but a discrepancy with the reference implementation cannot be excluded, and we will replace it if code becomes available. The tightest duration band retains only 7 negatives, and its role in our argument is corroborative, not primary. Downstream conditions use 10 seeds; at this power, differences smaller than roughly 0.05 in mean success rate are not resolvable, which is why we report SAL and jerk as indistinguishable from the baselines rather than as harmful. The oracle training condition has a smaller pool (716 vs. 1950) and is reported as a ceiling, not a matched comparison. Finally, our success-label experiments use machine-generated failures (can/mg); failure modes of human teleoperators may present differently, and extending the audit to human-failure pools is the clearest next step. We report can/paired only as a secondary setting because its matched-pair construction, in which each success/failure pair shares a near-identical trajectory prefix and diverges only in the final placement, concentrates the discriminating signal into a handful of terminal timesteps and simultaneously introduces an opposing length confound (successful demonstrations are systematically longer, length-alone AUROC 0.194); neither property is representative of the mixed-quality pools curation metrics are deployed on.

8 Conclusion

Published trajectory-smoothness curation metrics, evaluated exactly where their authors did not, on real operator labels under duration controls and on fixed-length data against task success, rank

episode duration more than demonstration quality. Once duration is controlled, the crude jerk baseline they were designed to supersede outperforms them, and the contact-aware metric inverts, and the same crossover reproduces on a second task without retuning. Training on the metrics’ own curated subsets shows the practical stakes are real but narrower than a five-seed pass suggested: curating by TED costs roughly a third of downstream success against random selection of the same size, a result robust to correction for multiple comparisons, while the other metrics are directionally below the baselines without being distinguishable from them. An oracle reaches 0.892 on the same pool, so the signal these metrics fail to find is unambiguously present in the data.

Reproducibility. All datasets are the public robomimic release [Mandlekar et al., 2021]; SAL follows its published formula and TED is reimplemented from the RINSE appendix as described in Section 3. Metric implementations, scoring scripts, and training configurations will be released at an anonymized repository upon acceptance and de-anonymized in the camera-ready or public posting.

Use of AI assistance. An AI assistant was used for code scaffolding, LaTeX drafting, and editing support during this project. All experiments were designed, executed, and verified by the author(s), and all reported numbers were produced by the released code and checked against the raw outputs.

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